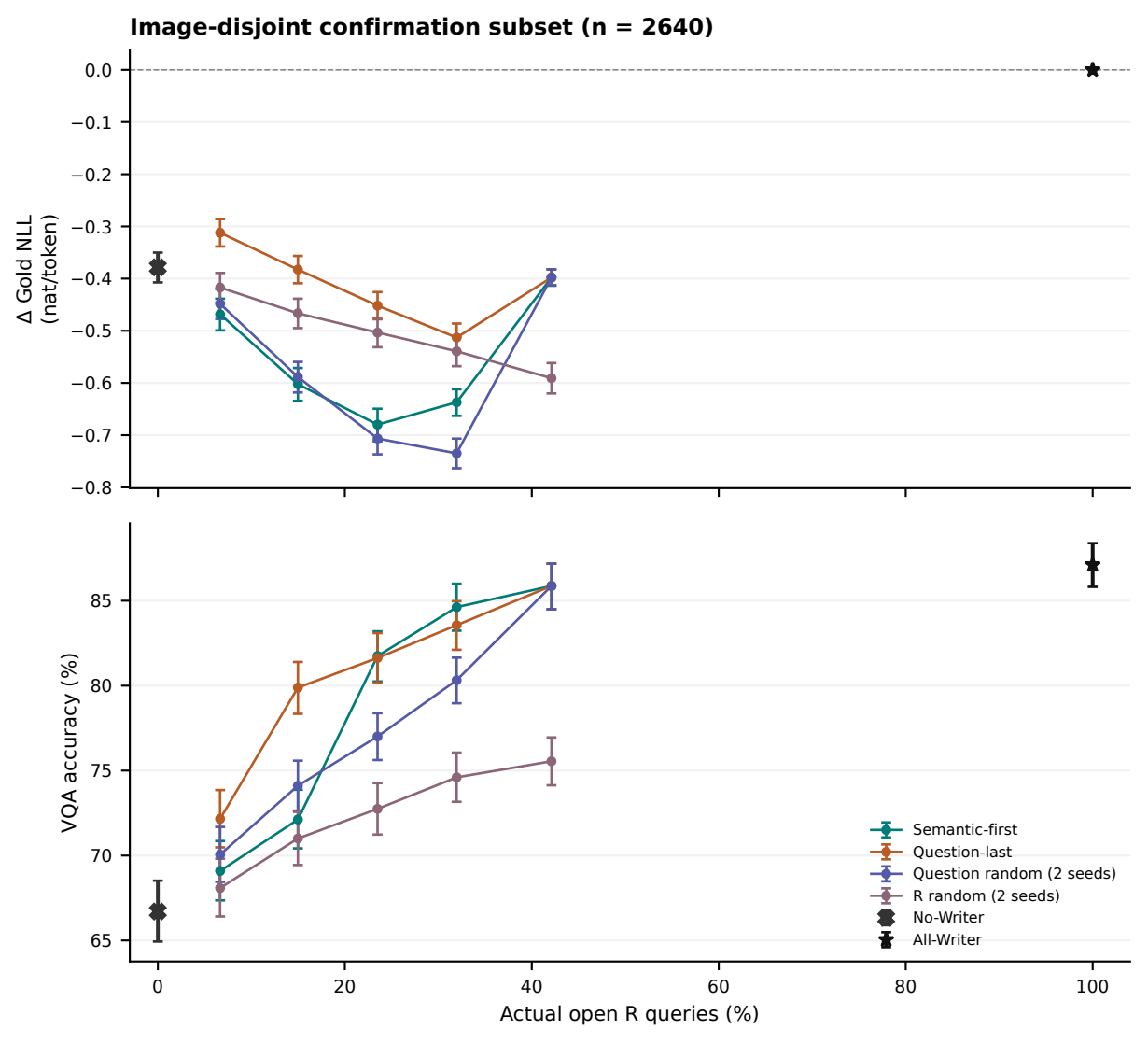
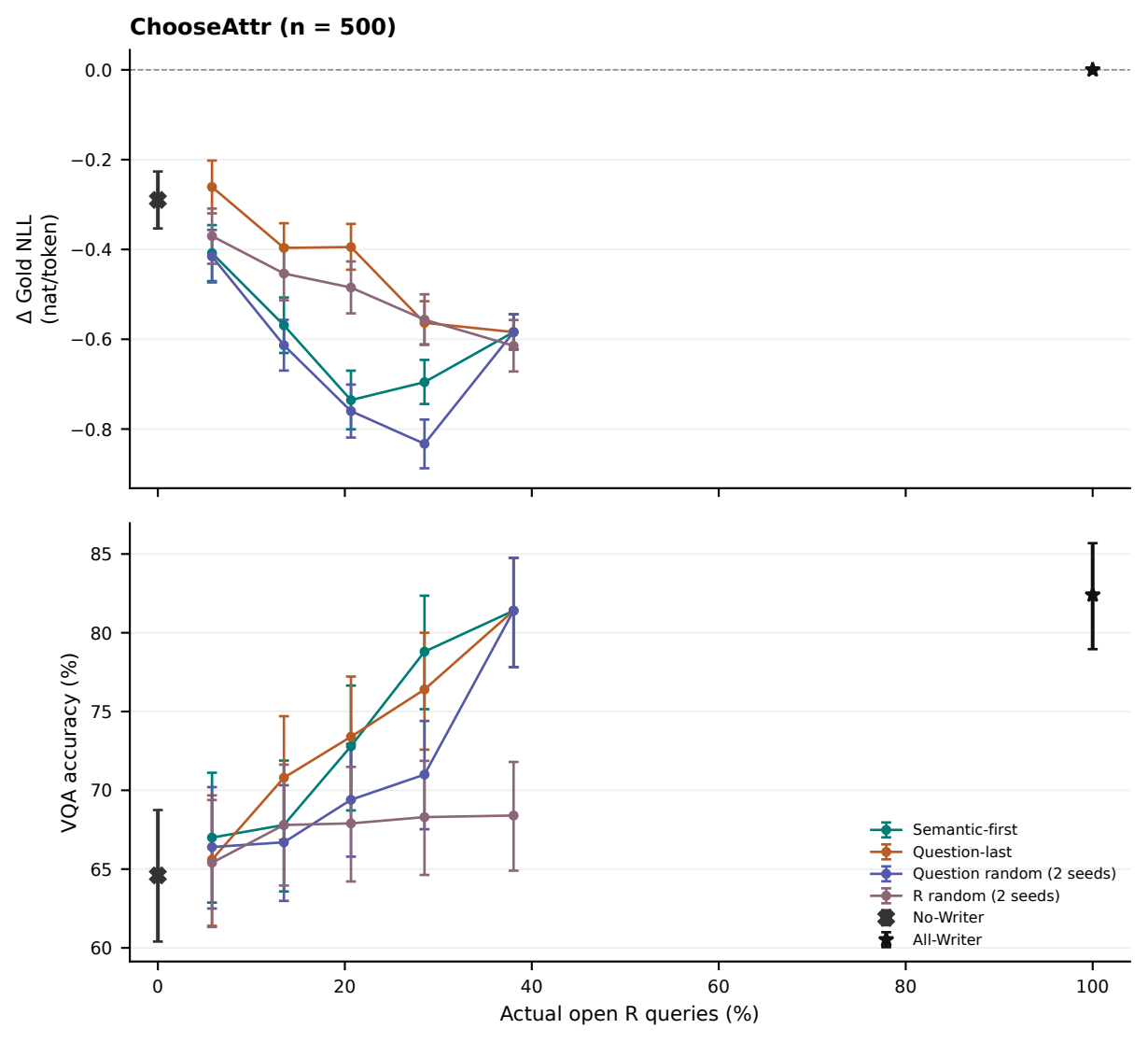
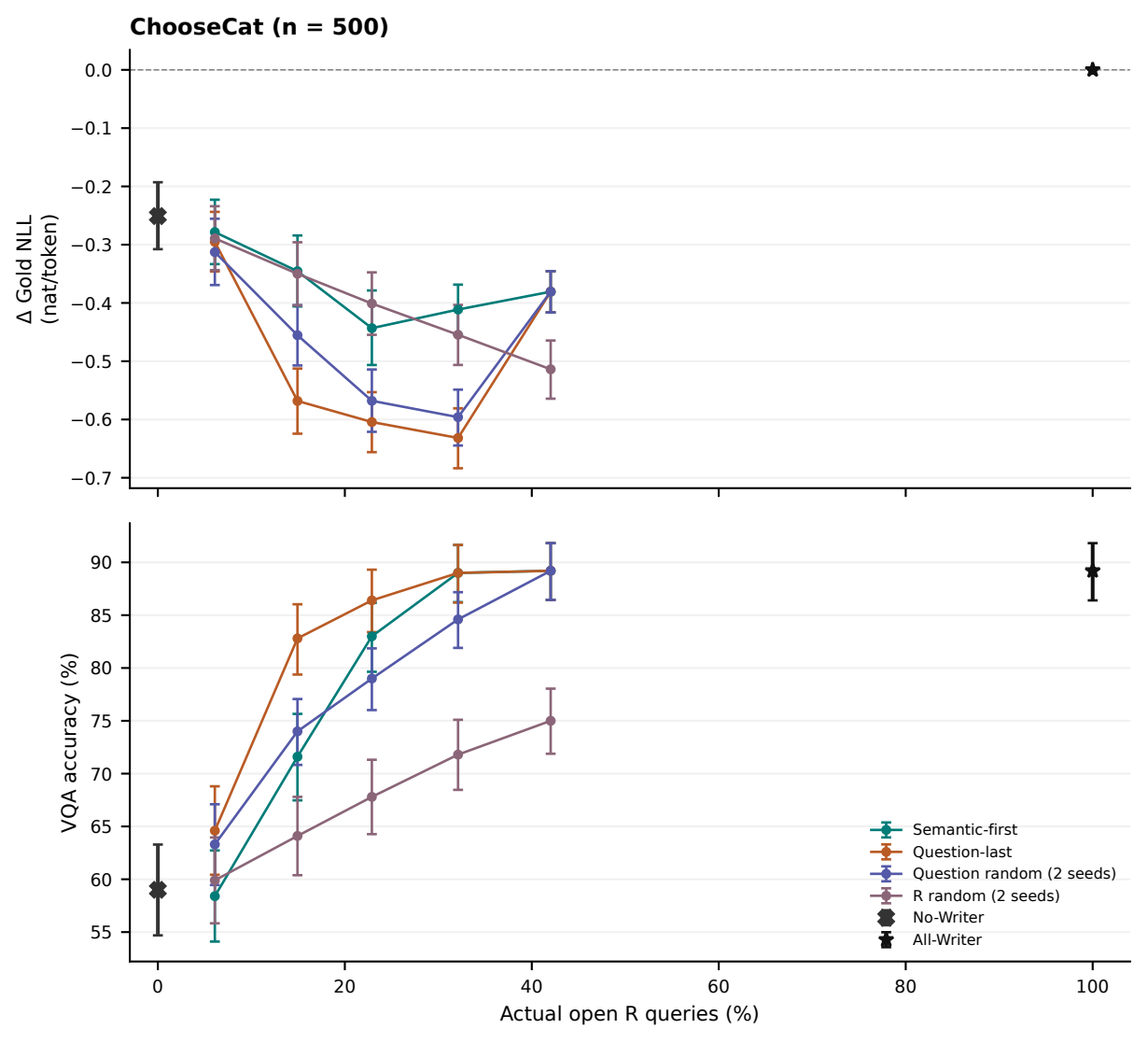
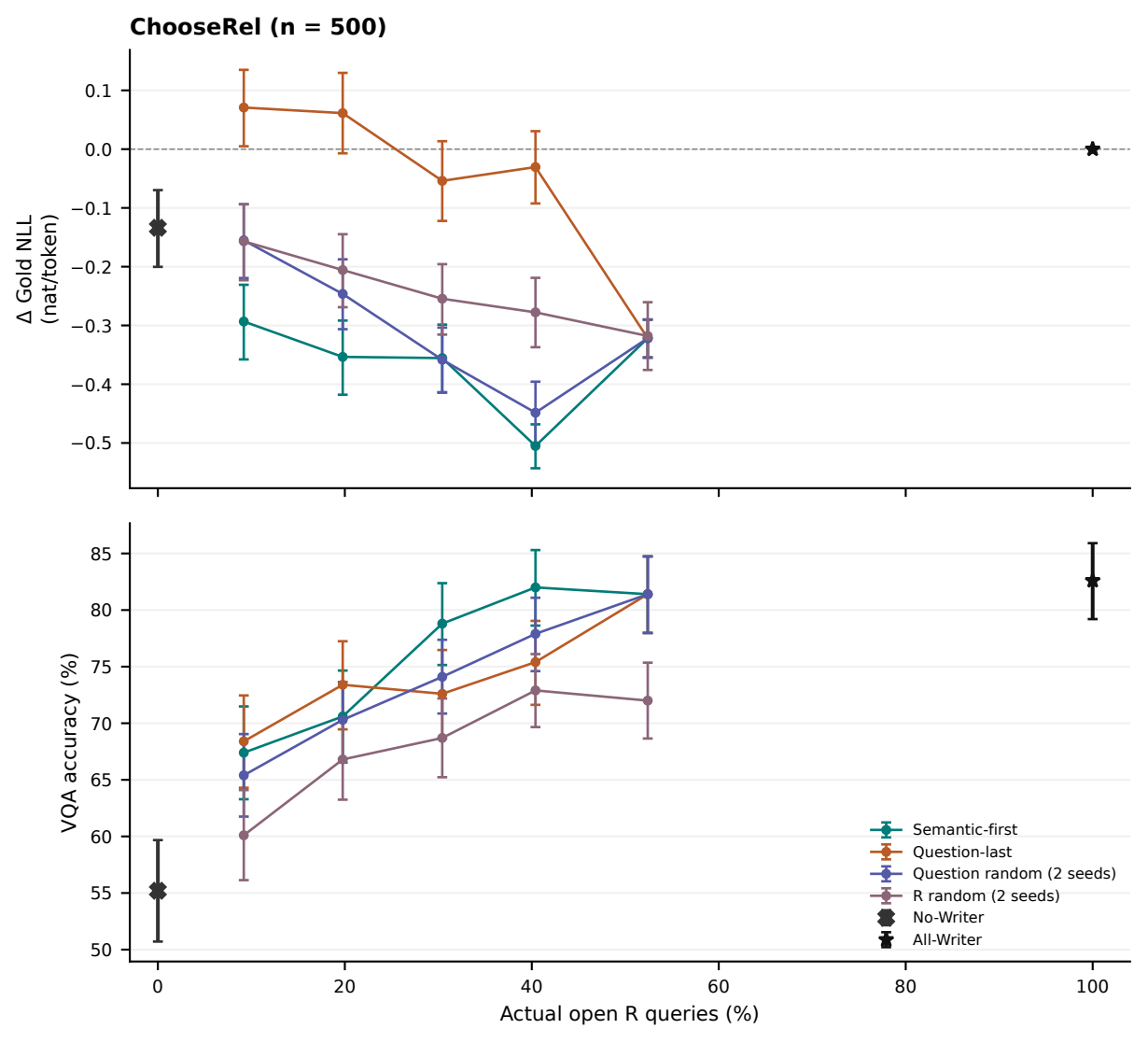
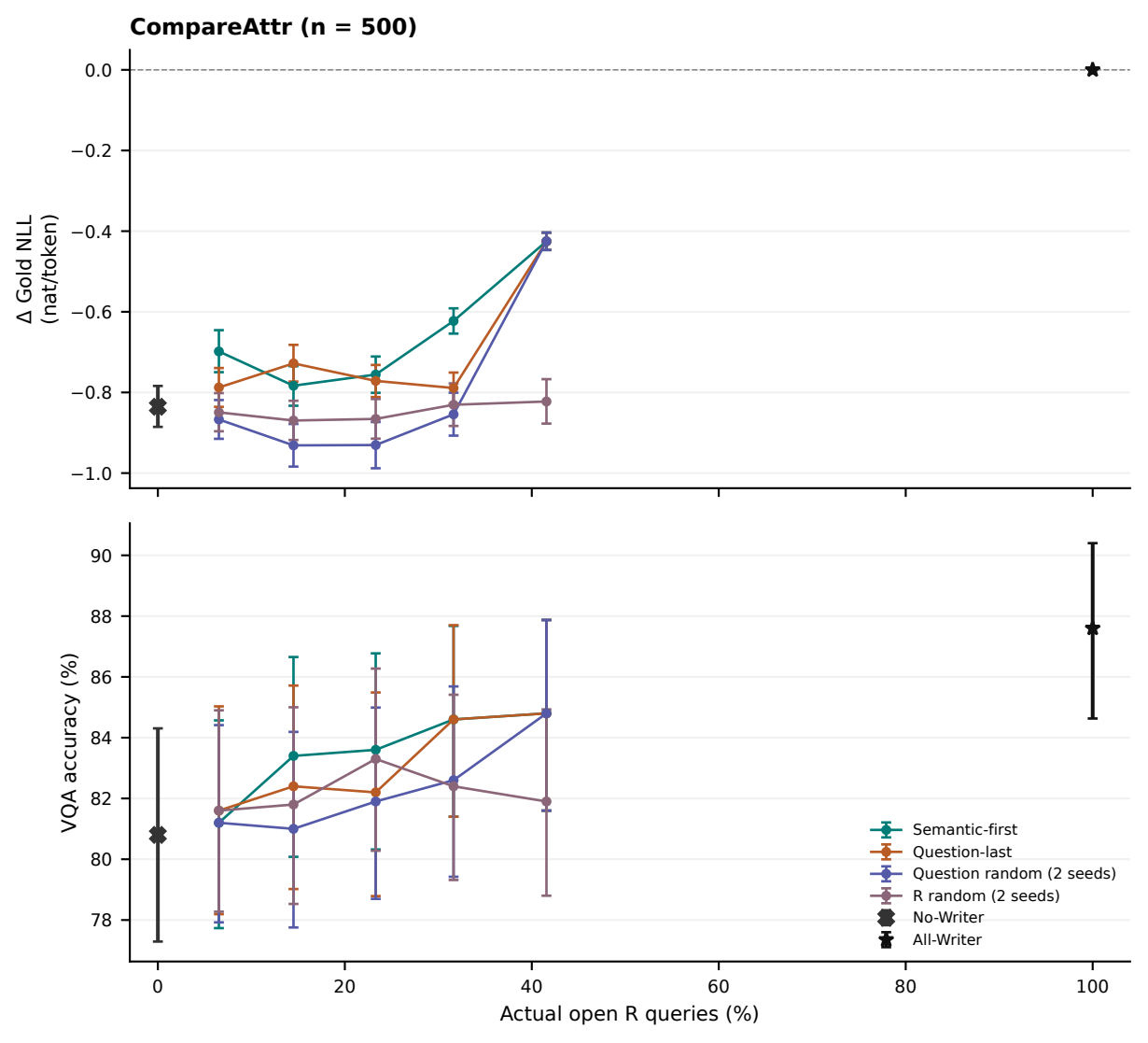
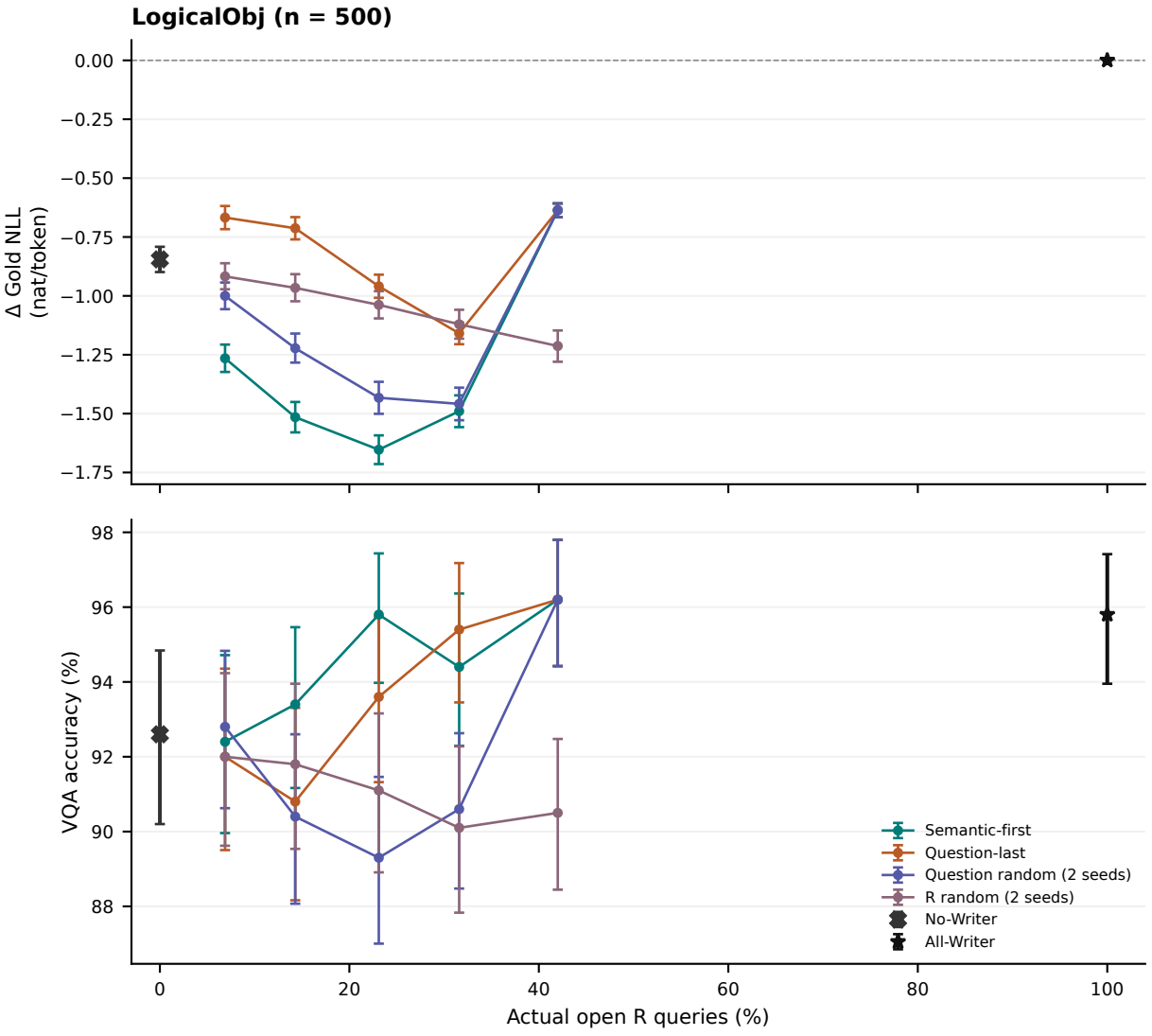

Image-disjoint confirmation subset (n = 2640)
Δ Gold NLL (nat/token)
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
−0.7
−0.8
VQA accuracy (%)
85
80
75
70
65
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer

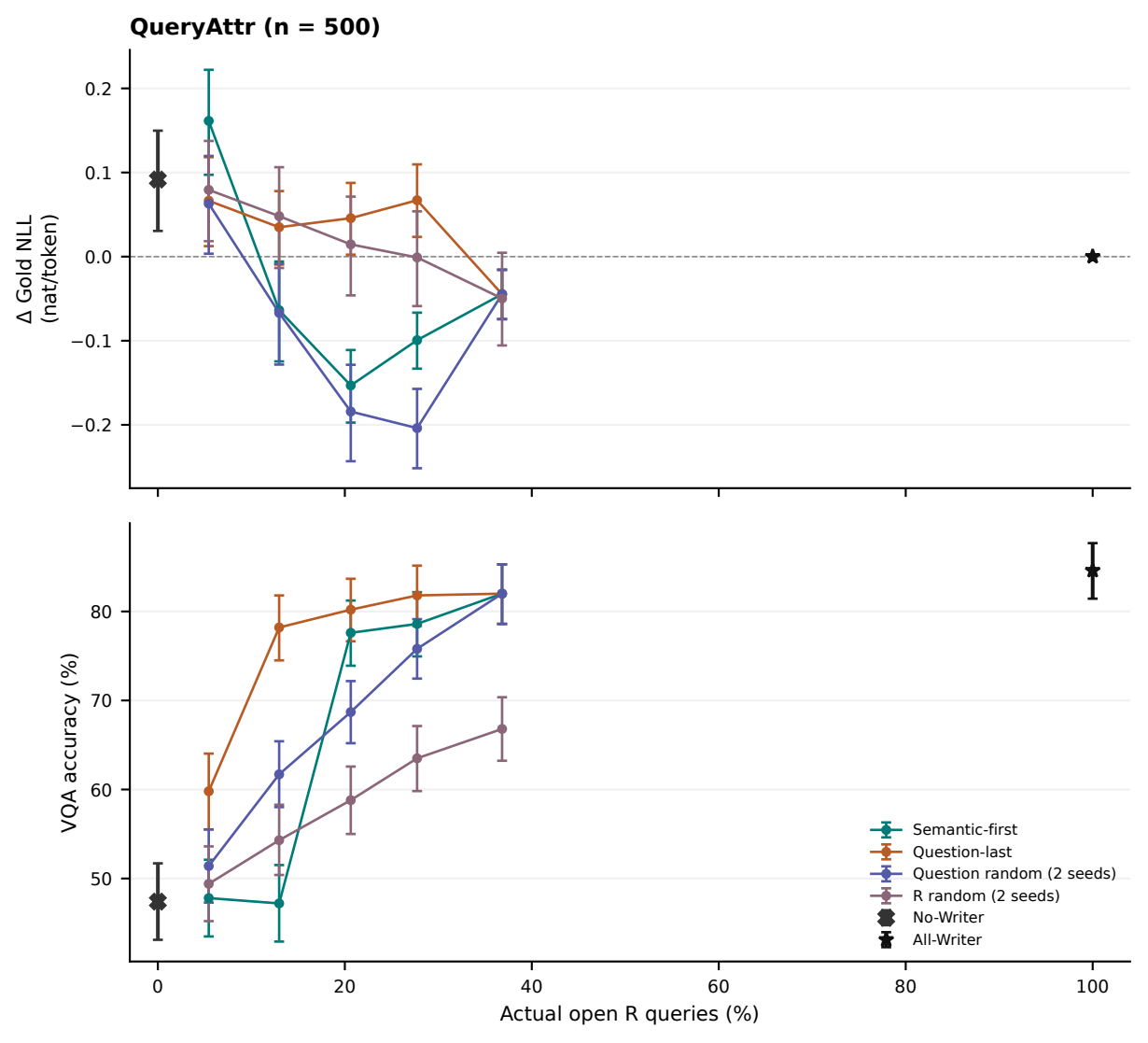

ChooseAttr (n = 500)
Δ Gold NLL (nat/token)
0.0
−0.2
−0.4
−0.6
−0.8
VQA accuracy (%)
85
80
75
70
65
60
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer

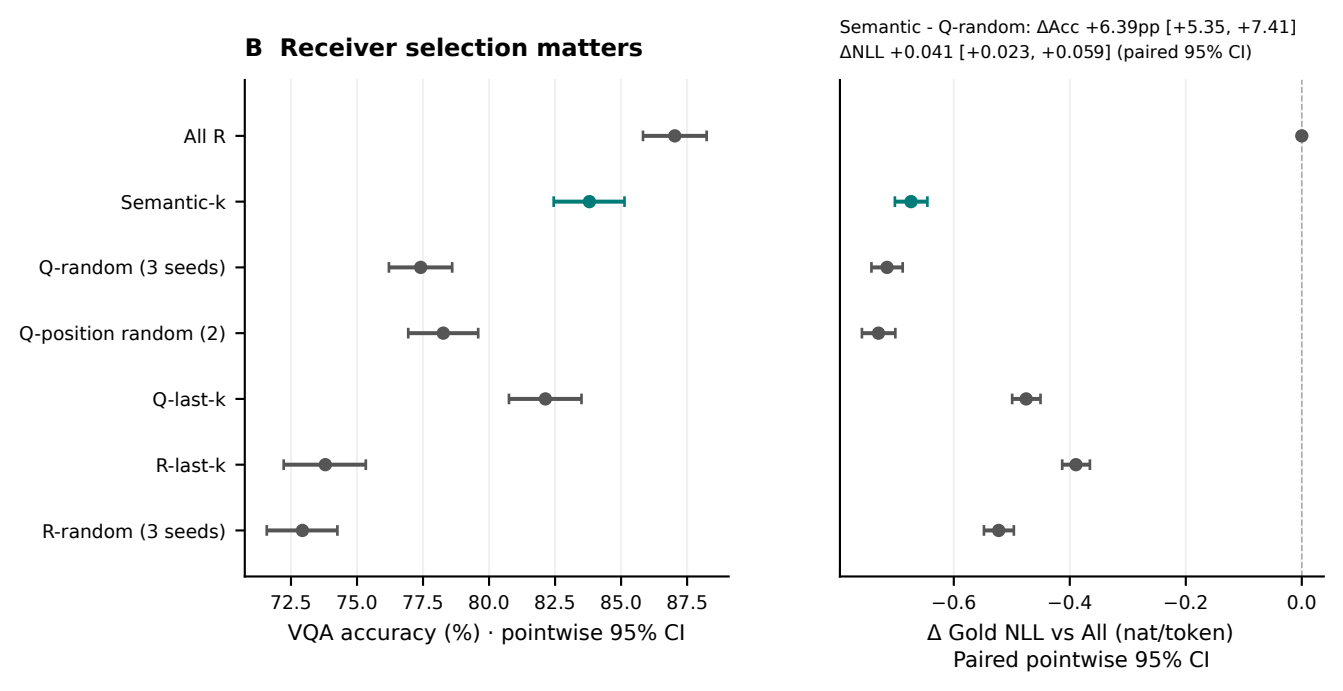

ChooseCat (n = 500)
Δ Gold NLL (nat/token)
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
−0.7
VQA accuracy (%)
90
85
80
75
70
65
60
55
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer


ChooseRel (n = 500)
Δ Gold NLL (nat/token)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
VQA accuracy (%)
85
80
75
70
65
60
55
50
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer


CompareAttr (n = 500)
Δ Gold NLL (nat/token)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
VQA accuracy (%)
90
88
86
84
82
80
78
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer


LogicalObj (n = 500)
Δ Gold NLL (nat/token)
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
VQA accuracy (%)
98
96
94
92
90
88
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer


QueryAttr (n = 500)
Δ Gold NLL (nat/token)
0.2
0.1
0.0
−0.1
−0.2
VQA accuracy (%)
80
70
60
50
0
20
40
60
80
100
Actual open R queries (%)
Semantic-first
Question-last
Question random (2 seeds)
R random (2 seeds)
No-Writer
All-Writer


B Receiver selection matters
All R
Semantic-k
Q-random (3 seeds)
Q-position random (2)
Q-last-k
R-last-k
R-random (3 seeds)
72.5
75.0
77.5
80.0
82.5
85.0
87.5
VQA accuracy (%) · pointwise 95% CI
Semantic - Q-random: ΔAcc +6.39pp [+5.35, +7.41]
ΔNLL +0.041 [+0.023, +0.059] (paired 95% CI)
−0.6
−0.4
−0.2
0.0
Δ Gold NLL vs All (nat/token)
Paired pointwise 95% CI